



Review Article

Hemocompatibility-related Adverse Events in Patients With Temporary Mechanical Circulatory Support: The Scoring Haemostasis Events and Assessment for Risk (SHEAR) Score



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Evaluation of treatment outcomes in patients supported by temporary mechanical circulatory support (tMCS) currently relies mainly on mortality, which may not sufficiently address other patient benefits or harms. Bleeding and thrombosis are major contributors to mortality. Still, current bleeding scores are not designed for critically ill patients undergoing tMCS, only consider selected populations, and do not account for the high heterogeneity among bleeding and thrombotic adverse events. To improve clinical management, a group of European experts has proposed a revised scoring system based on the MOMENTUM 3 Hemocompatibility Score and the Society of Cardiac Angiography and Interventions (SCAI) classification of cardiogenic shock. The new system termed the Scoring Haemostasis Events and Assessment for Risk (SHEAR) score, is divided into a baseline characterization stage and four escalating scoring stages encompassing all aspects of clinical relevance. This report summarizes the literature on hemocompatibility-related adverse events associated with tMCS, including bleeding, stroke, vascular access complications, hemolysis, thrombosis, and device failure. The SHEAR score provides a simple and rapid bedside scoring system aiming to provide a

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univocal tool to increase physician awareness of hemocompatibility complications at baseline and beyond, improve clinical research, and enable the capture of device-related complications that will inform relevant outcomes beyond mortality.

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Key Words: hemocompatibility; temporary mechanical circulatory support (tMCS); bleeding score; thrombosis; SHEAR; Impella; ECMO

THE CURRENT USE of survival alone to evaluate the clinical outcomes of patients supported by temporary mechanical circulatory support (tMCS)¹ appears inadequate.

Hemocompatibility, the constellation of synergistic and/or parallel bleeding and thrombotic events that ensue in response to a material or device make coagulation management challenging. Moreover, hemocompatibility-related events severely jeopardize outcomes and are the main causes of morbidity and mortality in patients on tMCS.² To improve outcomes in this critically ill population and evaluate the use of tMCS, an improved scoring system for the specific assessment of hemocompatibility in critically ill tMCS-supported patients is urgently needed.

To establish a new scoring system, several issues must be navigated. First, there is substantial heterogeneity among definitions of bleeding and thrombotic events that prevent accurate stratification of severity, potentially undermining the ability to define the issues at hand and collate and draw meaningful conclusions from the available data. Second, many of the available bleeding scores have been designed for selected, stable study populations, such as after cardiac surgery, in the context of dual antiplatelet therapy, and percutaneous coronary intervention³; so far, no scoring systems are designed to assess the critically ill being treated with tMCS. Third, a major step in improving outcomes of tMCS treatment of cardiogenic shock (CS) is finding the optimal balance between bleeding and thrombosis. It is, therefore, crucial to systematically identify these events and carefully evaluate the real value of the device itself in clinical studies.⁴

This is complicated by the variety of hemocompatibility-related events that can occur. For example, a single patient can suffer multiple events of low severity, a single severe and/or life-threatening complication, or both bleeding and thrombosis with heterogeneous clinical impacts. Likewise, some events may lead to long-lasting functional deficits, while others are only relevant for acute management. The definition and stratification of these adverse events (AEs) their assessment at baseline and during treatment, and the characteristics of the MCS device are pivotal to a more refined hemocompatibility scoring system.

Methods

From this perspective, a group of European experts has leveraged the strengths of the MOMENTUM 3¹ hemocompatibility score and the Society of Cardiac Angiography and Interventions (SCAI) classification of CS⁵ to outline a consensus definition and stratification of thrombotic and bleeding complications in patients supported with tMCS, termed the Scoring Haemostasis Events and Assessment for Risk

(SHEAR) score. The definition encompasses all aspects of clinical relevance and device-specific features.

The development group, a multidisciplinary panel including cardiac surgeons, intensivists, and cardiologists, was identified by the first author based on individual expertise and experience with MCS. In 2022, the expert group convened an initial meeting and agreed on potentially relevant aspects derived from a combination of prior studies and expert opinion. Then, two in-person and three online meetings with 13 participants were conducted.

A literature search was conducted, including reviews, guideline documents, and clinical studies on adult patients receiving MCS, published in English in PubMed, Embase, and Cochrane databases. The search strategy is reported in the Supplementary Material.

A series of virtual and in-person meetings were conducted to propose the definitions, which were then assessed for tier assignment by voting. The purpose of the proposed SHEAR classification of hemocompatibility-related AEs is to: (1) provide a simple method to increase the physician's knowledge base and stratify hemostasis complication severity, (2) provide a tool to stimulate and improve clinical research in this area, and (3) enable the capture of differences between device-based therapies and how they inform relevant outcomes beyond mortality. In the long term, the latter will crucially inform treatment sequencing strategy. To achieve these ends, a consistent approach to hemocompatibility-related complications data collection and event severity adjudication are essential. Patient-related risk factors at baseline will be identified by the clinician, ensuring they are incorporated during the care trajectory. The baseline methodology is constructed along the same lines as the globally endorsed SCAI classification for CS, applied in similar clinical scenarios by healthcare providers and researchers worldwide, making its application easier and more practical.

In the proposed SHEAR algorithm, the net burden of hemocompatibility-related AEs (HRAEs) can be assessed using a single-stage baseline score and a four-stage escalating score system of hierarchical severity steps derived from the total score of events encountered during the entire duration of tMCS. Definitions and stages consist of laboratory parameters, tMCS device data, and clinical events, and each definition incorporates a different combination of these data, providing physicians with a quick, intuitive bedside ranking in a specific category of severity.

Heterogeneity of Bleeding and Thrombosis Definitions Across Trials

There is a misconception that the wide variability of bleeding and thrombotic complications reported in clinical studies

on CS and tMCS are solely related to the device.⁴ It is important to stress that these complications might be due to several factors, including differences in patient characteristics, concomitant therapies, timing of event reporting, definitions, and the specific device used. For example, recording hemoglobin decreases without clinically overt signs of bleeding may trigger the clinician's awareness of hemolysis produced by the artificial pump and of the multiple mechanisms producing loss of hemoglobin in critically ill patients (eg, repetitive blood sampling or bone marrow suppression due to critical illness). The timing of events is also important. Different temporal relationships and specific time intervals have been applied in different clinical settings; for example, coronary artery bypass graft (CABG)-related, Bleeding Academic Research Consortium (BARC) type 4⁶ is performed while a punctual intervention (surgery, thrombolysis) is also performed. In the SHEAR scoring system, this is overcome by the extended and variable duration of tMCS support, which will be definitively linked to a hemocompatibility event. Another factor is the patient population under study. Some randomized controlled trials (RCTs), such as the ECMO-CS,⁷ report different rates of AEs according to either intention-to-treat or per treatment data analysis. Despite these population differences, patients with CS have a higher baseline risk of bleeding even without the use of tMCS.⁸

Regardless of the definition, several studies⁹ have demonstrated that bleeding and thrombosis are associated with an increased risk for short- and long-term adverse outcomes. The exact mechanisms underlying this relationship are unknown but may include antithrombotic or anticoagulant medications, systemic inflammatory response syndrome (SIRS), transfusions, pre-existing comorbidities, and the mechanical interaction between the pump and the native cardiovascular system (eg, foreign body blood contact and shear stress).

This clinical situation is further complicated by the lack of validation of several blood tests in critically ill cardiac patients. Notably, the consequence of hemostatic derangement could also lead to the cessation of evidence-based therapies, such as dual antiplatelet therapy.

The relationships between severe AEs are closely intertwined because management to mitigate one may directly promote the other. For example, withholding anticoagulants controls bleeding but can promote thrombosis and a feedback effect between these types of AEs. Historically, these events have been reported separately; however, understanding their interplay and the accumulation of events in each patient offers a new and more complete perspective, albeit with additional challenges. Therefore, there is a distinct potential advantage to improving patient prognosis by evaluating hemocompatibility-related outcomes as a consolidated cumulative endpoint.

Available Definitions of Severe AEs Associated With tMCS

AEs associated with tMCS and/or AEs on tMCS

Hemocompatibility-related complications associated with tMCS can include hemolysis, thrombotic events (eg, stroke, systemic embolism, pump thrombosis), hemorrhagic

complications (eg, access site bleeding, intracranial hemorrhage, gastrointestinal bleeding), arrhythmias, and device failure. To date, no internationally accepted consensus exists on which classification should be used to describe these AEs or provide guidance on their occurrence, severity, and potential consequences that may counterbalance the positive effects of tMCS in CS patients.¹⁰

A consensus-based recommendation for primary clinical trial outcomes evaluating hemostatic blood products and agents in patients with bleeding has been developed and included cardiac surgical and MCS patients.¹¹ The consensus group agreed upon identifying relevant thromboembolic and bleeding events.

The current work, however, introduces a staged classification of thrombotic and/or bleeding events and, different to currently available and common classifications for the most frequent AEs in clinical practice ([Supplementary Material](#)), it applies to specific scenarios of CS and tMCS.

AE: Bleeding

Bleeding is the most frequently reported AE in large-scale RCTs and observational studies (data available from all RCTs and from 14 of 16 observational studies). It is often defined based on the BARC classification into five stages.⁶

Two easier, more pragmatic approaches consisting of three stages (minor, moderate, or severe and/or life-threatening bleeding) are the Global Use of Streptokinase and t-PA for Occluded Coronary Arteries (GUSTO)¹² and the Thrombolysis in Myocardial Infarction (TIMI) definitions.¹³ These have been used in RCTs on myocardial reperfusion with fibrinolysis ([Tables 1 and 2](#)). Other classifications are used in large registries such as the Extracorporeal Life Support Organization (ELSO) Registry,¹⁴ the Society for Thoracic Surgeons Adult Cardiac Surgery Database (ACSD) registry¹⁵ or the American College of Cardiology-National Cardiovascular Data Registry.¹⁶

Here, bleeding is frequently only reported as severe (defined by a need for intervention or transfusion) compared with non-severe events ([Table 3](#)). Only a few observational studies have used a validated definition (3 of 16 used the GUSTO definition, 1 of 16 the Valve Academic Research Consortium-2 [VARC-2],¹⁷ and 2 of 16 the STS ACSD classifications), and none used BARC or TIMI. Moreover, the transfusion rate or the number of blood products transfused (packed red blood cells [pRBCs] or fresh frozen plasma), which can be surrogates of bleeding severity, are rarely reported (1 of 16 observational trials and 2 of 7 RCTs).

AE: Stroke

Stroke as an AE is defined as ischemic stroke, intracerebral bleeding, or hemorrhagic stroke confirmed by computed tomography or magnetic resonance imaging; strokes are often well recognized and analyzed in CS studies. Although recognized scores such as the Rankin and modified Rankin Scale scores are available and previously used for left ventricular assist device trials such as the MOMENTUM 3 trial, stroke

Table 1
Cardiogenic Shock Randomized Controlled Trial Endpoints

NCT Number Acronym	Bleeding, Thrombosis	Vascular Complications	Stroke	Embolism, Hemolysis	Device Failure	Structural or Myocardial Injuries	Transfusion	MI/Stent Thrombosis /Revascularization
Acute MCS as primary intervention in CS patients								
NCT00491036 IABP-SHOCK2	GUSTO, BARC, TIMI, NR	Peripheral ischemic vascular complications requiring surgical or interventional therapy	Stroke with any new neurologic symptoms in conjunction with signs of ischemia or bleeding in cranial computed tomography	NR, NR	NR	NR	NR	NR
NCT04369573 ALTSHOCK-2	BARC >3, NR	Reported without definition	NR	Reported without definition, NR	NR	NR	NR	NR
NCT03813134 EURO SHOCK	BARC 3-5 at discharge and 12 mo, NR	VARC-2 definition	Acute neurologic deficit lasting longer than 24 h at discharge and 12 mo	NR, NR	NR	NR	NR	Recurrent MI at discharge and 12 mo
NCT04184635 ANCHOR	Major bleeding at 30 d (TIMI definition), ECMO-related complications at 30 d	ECMO-related complications at 30 d (infection at ECMO cannulation sites requiring antibiotics, hemorrhage, limb ischemia requiring surgery, cannula or circuit thrombosis, overt pulmonary edema, thrombocytopenia, gaseous emboli, and hemolysis)	Any new neurologic symptoms in association with signs of ischemia or hemorrhage in a cranial CT or MRI at 30 d	ECMO-related complications at 30 d, ECMO- related complications at 30 d	NR	NR	Number of pRC at 30 d	Recurrent MI and recurrent revascularization (PCI or CABG)
NCT01633502 DANGERSHOCK	GUSTO, NR	Limb ischemia, need for surgical intervention or amputation	NR	Reported without definition, significant hemolysis	Device failure	Damage to Ao valve	NR	NR
NCT03637205 ECLS-SHOCK	NR, NR	NR	CPC at 30 d, 6 and 12 mo	NR, NR	NR	NR	NR	Recurrent MI at 30 d, 6 and 12 mo
NCT02301819 ECMO-CS	Clinically relevant bleeding, NR	Leg ischemia	CPC at 30 d	NR, NR	Technical complications	NR		NR
NCT05366452 ULYSS	BARC $\geq$ 3, NR	Number of vascular complications requiring medical or surgical care	NR	Number of systemic embolisms, clinical hemolysis	Requiring extraction	NR	number of pRCs	MI at discharge, 30 d, and 1 y

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Table 1 (continued)

NCT Number Acronym	Bleeding, Thrombosis	Vascular Complications	Stroke	Embolism, Hemolysis	Device Failure	Structural or Myocardial Injuries	Transfusion	MI/Stent Thrombosis /Revascularization
NTR3450 IMPRESS	Bleeding with hemoglobin drop ≥5 g/dL and/or bleeding necessitating ≥2 packed cells of blood product transfusion and/or need for surgery to control bleeding, NR	Major bleed arising at arterial access site requiring extraction of device, thrombotic occlusion of femoral artery, limb ischemia requiring extraction of device, need for vascular surgery to correct a vascular complication	Any stroke confirmed by neurologist + CT scan	NR, any hemolysis requiring replacement or extraction	Any device failure requiring replacement or extraction	NR	NR	MI
NCT05506449 RECOVER IV	SHARC types 3-5, NR	Minor and major vascular complications (SHARC definition)	Any stroke at 30 d and/or Rankin score at 30 d, 6 mo, and 1 y and/or 30 d survival with mRS score ≥3 and/or number of participants with NeuroARC Type 1 stroke	NR, minor and major hemolysis	NR	NR	NR	Repeat target vessel revascularization at 30 d, 6 mo, and 1 y
NCT05106491 PULSE	Severe bleeding, NR	NR	NR	NR, severe hemolysis	Device-related complications	NR	NR	NR
CS RCTs with acute MCS as associated intervention								
NCT02754193 HYPOECMO	GUSTO, NR	NR	Rate at 30, 90, and 180 d	NR, NR	NR	NR	Number of pRCs	NR
NCT05527717 RESCUE-SHOCK	Rates of bleeding and severe bleeding (BARC) at 90 d and 12 mo, NR	Rates of critical limb ischemia at 30 d	CPC score 3-5 at 30 d	NR, NR	NR	NR	NR	Rate of MI and rate of stent thrombosis at 3 and 12 mo
NCT04876040 ISO-SHOCK	NR, NR	NR	NR	NR, NR	NR	NR	NR	NR
CS RCTs without acute MCS								
NCT01927549 CULPRITSHOCK	GUSTO, BARC, TIMI, NR	NR	NR	NR, NR	NR	NR	NR	Recurrent MI and revascularization (PCI or CABG) at 30 d and 12 mo
NCT03207165 DOREMI	NR, NR	NR	In-hospital rate of TIA or stroke	NA, NR	NR	NR	NR	Nonfatal MI during hospitalization
NCT03773822 COCCA	Rate of bleeding at 28 and 90 d, NR	NR	Rate of stroke at 28 and 90 d	NA, NR	NR	NR	NR	NR

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Table 1 (continued)

NCT Number Acronym	Bleeding, Thrombosis	Vascular Complications	Stroke	Embolism, Hemolysis	Device Failure	Structural or Myocardial Injuries	Transfusion	MI/Stent Thrombosis /Revascularization
NCT04020263 LevoHeartShock	NR, NR	NR	Rate of recurrent MI and urgent coronary revascularization in a composite endpoint (D 30, 60, 90, 180, and 12 mo)	NR, NR	NR	NR	NR	Recurrent MI and urgent coronary revascularization in a composite endpoint (D 30, 60, 90, 180, and 12 mo)
NCT05168462 NORSHOCK	NR, NR	NR	NR	NR, NR	NR	NR	NR	Infarct size measured by hs-Troponin T at 0, 6, 12, 24, 36, and 72 h
NCT05485376 PACCS	NR, NR	NR	NR	NR, NR	NR	NR	NR	NR
NCT05298124 MINOS	Rate of bleeding at 90 d, any intracranial bleeding (excluding microhemorrhages <10 mm evident only on gradient- echo MRI), clinically overt signs of hemorrhage associated with a drop in hemoglobin of ≥5 g/dL or a ≥15% absolute decrease in hematocrit, fatal bleeding (bleeding that directly results in death within 7 d), NR	Rate of vascular complications at 90 d (access site–related arterial or venous injury or injury to surrounding structures)	Rate of stroke or TIA at 90 d	NR, NR	Device success	Rate of structural heart complications at 90 d (cardiac perforation or pseudoaneurysm)	NR	NR
NCT05594342 IVA-CS	NR, NR	NR	NR	NR, NR	NR	NR	NR	NR
NCT03551964 DAPT-SHOCK- AMI	BARC ≥3 at 30 d, NR	NR	Rate of stroke in a composite secondary endpoint	NR, NR	NR	NR	NR	Definite stent thrombosis at 30 d
NCT05267886 DOREMI-2	NR, NR	NR	In-hospital rate of TIA or stroke	NR, NR	NR	NR	NR	Nonfatal MI during hospitalization

NOTE. Only interventional, multicentric RCTs on CS (preshock and stage B and A SCAI are not included) registered on Clinicaltrial.gov in May 2023 were included. Trials on early RRT under VA-ECMO in CS patients were not included. Trials addressing left ventricular unloading under VA-ECMO were not included.

Abbreviations: aMCS, acute mechanical circulatory support; Ao, aortic; BARC, Bleeding Academic Research Consortium; CABG, coronary artery bypass graft; CPC, cerebral performance category; CS, cardiogenic shock; CT, computed tomography; DAPT, dual antiplatelet therapy; ECMO, extracorporeal membrane oxygenation; GUSTO, Global Use of Streptokinase and t-PA for Occluded Coronary Arteries; hs-Troponin T, high sensitivity Troponin T; LV, left ventricle; MCS, mechanical circulatory support; MI, myocardial infarction; MRI, magnetic resonance imaging; mRS, modified Rankin scale; MV, multivessel; NA, not available; NCT, national clinical trial; NeuroARC, Neurological Academic Research Council; NR, not reported; PCI, percutaneous coronary intervention; pRC, packed red blood cells; RCT, randomized

Table 2
Adverse Events in Randomized Clinical Trial Cardiogenic Shock Studies

NCT Number Trial Acronym Reference	Timing of Primary Endpoint	aMCS Type (n)	Bleeding/ Severe Bleeding (%)	Transfusion (%), Number of Transfusions	Reported AE (%) CPC ≥3	Recurrent MI (%)	Stent Thrombosis (%)	Device Failure (%)
NCT00491036 IABP- SHOCK2 Thiele, 2012	In-hospital	IABP (300)	17.3/3.3	NR, NR	LB1: 4.3 STK: 0.7 CPC: NR	3	1.3	NR
		Control (298)	16.4/4.4		LB1: 3.4 STK: 1.7 CPC: NR	1.3	1	
NCT03813134 EUROSHOCK Banning, 2023	30 d	ECMO (17)	NR/36.6	NR, NR	LB1: 21 STK: 0 CPC: NR	0	NR	NR
		Control (18)			LB1: 0 STK: 11 CPC: NR	11		
NCT02301819 ECMO- CS* Ostadal, 2023	30 d	Immediate ECMO (81)	30.9/NR	NR, NR	LB1: 9.9 STK: 3.7 CPC: 48.3	NR	NR	1.7
		Control (36)	13.9/NR		LB1:8.3 STK: 0.0 CPC: 45.8			0
NTR3450 IMPRESS Ouweneel, 2017	30 d (sAE reported at 6 mo)	Impella CP (24)	NR/33	46, 6 pRBCs/3 fFP	LB1: 4.0 HLS: 8.0 STK: 4.0 CPC: NR	4	NR	0
		IABP (24)	NR/8	33, 3 pRBCs/0 fFP	LB1: 0.0 HLS: 0.0 STK: 4.0 CPC: NR	8		0
DANGER	180-d mortality	Impella CP (179)	21.8	NR	LB1: 5.6	NR	NR	1.2
		Standard therapy (176)	11.9	NR	LB1: 1.1	NR	NR	33.3 (Other device)
CS RCTs with aMCS as associated intervention								
NCT02754193 HYPOECMO Levy, 2022	30 d	Moderate hypothermia (167)	41	32, 8.3 pRBCs	NR CPC: NR	NR	NR	NR
		Normothermia (166)	42	30, 5.4 pRBCs				
CS RCT without aMCS								
NCT01927549 CULPRITSHOCK Thiele, 2017	30 d	Culprit only PCI (344)	16.6/12.5	NR, NR	STK: 3.5 CPC: NR	1.2	NR	NR
		Multivessel PCI (341)	22/15.3		STK: 2.9 CPC: NR	0.9		
NCT03207165 DOREMI Mathew, 2021	30 d	Milrinone (96)	NR/NR	NR, NR	STK: 1.0 CPC: NR	1	NR	NR
		Dobutamine (96)			STK: 2.0 CPC: NR	0		

NOTE. None of the studies reported other vascular events, thrombosis, systemic embolism, structural myocardial injuries, or urgent revascularization events. Abbreviations: AE, adverse event; aMCS, acute mechanical circulatory support; CPC, cerebral performance category; CS, cardiogenic shock; ECMO, extracorporeal membrane oxygenation; fFP, fresh frozen plasma; HLS, hemolysis; IABP, intra-aortic balloon pump; LB1, limb ischemia; MCS, mechanical circulatory support; MI, myocardial infarction; NCT, clinical trial identifier; NR, not reported; PCI, percutaneous coronary intervention; pRBCs, packed red blood cells; RCT, randomized controlled trial; sAE, serious adverse event; STK, stroke.

severity was never reported, which indicates another weakness in this area.¹⁸

AE: Vascular

Despite its frequent occurrence, other AEs associated with tMCS, such as vascular complications and limb ischemia, are poorly defined. A consensus definition for vascular complications is available from the VARC-2, classifying complications as minor and major,¹⁶ but this score is rarely used and remains under-reported (one RCT). Studies predominantly report vascular complications only when interventional or surgical treatment or amputation is required (Tables 1-4). Overall, vascular

complications and acute limb ischemia are more frequently reported in RCTs (4 of 5) whereas they are less frequently collected in observational studies (6 of 16).

AE: Hemolysis

Hemolysis is usually defined based on an increase of lactate dehydrogenase and/or free hemoglobin and can potentially trigger device explantation.¹⁰ However, sometimes hemolysis is only reported based on clinicians' judgment,¹⁸⁻²⁰ which, albeit significant in daily clinical practice, may significantly limit its objectivity. Despite the relevant biologic impact of free hemoglobin on shock hemodynamics,^{21,22} including acute

Table 3
Adverse Events in Observational Cardiogenic Shock

Author, Date Site (Type) Country Time Period	Type of CS (n)	Timing for Primary Endpoint	aMCS Type (n) Survival (%)	Bleeding (%)/ Severe Bleeding (%) Definition of Bleeding	Transfusion (%)/Number	Limb Ischemia (%)/Severe Limb Ischemia (%) Definition	Other Vascular Complications (%) Definition*
Danial, 2023 Monocentric (retrospective) France 2015-2018	All (1,253 /233 AMICS)	In-hospital	ECMO (1,253) 11.1-73.3 according to etiology of CS	25/NR General bleeding included hemothorax, hemopericardium, and gastrointestinal	NR/NR	9.0/0.9 Requiring surgery or amputation	12.0 NR
Mansour, 2022 Multicentric (retrospective) France 2020-2022	All but COVID associated (620)	In-hospital	ECMO (52 VA-ECMO/ 568 VV-ECMO) 44.3	48.4 VV-ECMO, 59.6 VA- ECMO/NR Bleeding events included intracranial bleeding, upper or lower gastrointestinal hemorrhage, peripheral cannulation site bleeding, retroperitoneal bleeding, and pulmonary hemorrhage	NR/4 rRBCs/0 fFP	1.1 VV-ECMO, 13.5 VA ECMO/NR NR	NR NR
Schrage, 2023 Multicentric (retrospective) EU and US NR	All (421 /238 AMICS)	In-hospital	ECMO + Impella (421) 41.8 (30 d)	41.8/36.8 GUSTO	NR/NR	19.7/NR NR	NR NR
Lee, 2021 Multicentric (retrospective) Korea 2014-2018	All (362/AMICS 207)	30 d	ECMO (362 (11.6% with associated IABP) 59.1	GI bleeding 3.3, ECMO site bleeding 12.4/NR NR	NR/NR	7.7/NR NR	NR NR
Acharia, 2020 Multicentric (retrospective) ELSO Registry 2014-2018	All (6,646/AMICS 665)	30 d	ECMO (6,646) AMICS 40.2 v non-AMI 42.1	GI bleeding 5.5, ECMO site bleeding 26.1%/NR Hemorrhagic complications requiring >3 pRBCs/calendar day in adults or other intervention such as surgical or endoscopic intervention	NR/NR	6.5/3.8 Requiring amputation and/or fasciotomy	NR NR
Schrage, 2023 Multicentric (retrospective) EU and US 2016-2021	Non-MI (890)	30 d	aMCS (386/42% Impella, 35% ECMO, and 23% ECMELLA) 49.1 with v 54.2 without MCS No MCS (504) As above	56.9/16.5 GUSTO 23.6/6.4 GUSTO	NR/NR	NR/6.7 Requiring intervention	NR NR
Varshney, 2023 Multicentric (retrospective) EU 2014-2019	ADHF bridge to LVAD or HTx (8,777/2,778 HTx and 5,933 LVAD)	30 d postoperative	Medical (5,013) 7.0 IABP (20,816) 9.0 pMCS (417) 10.0 ECMO (465) 22.0	NR/19.0 STS ACS definitions NR/2 2.0 STS ACS definitions NR/23.0 STS ACS definitions NR/36.0 STS ACS definitions	NR/NR	NR/NR NR	NR NR

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Table 3 (continued)

Author, Date Site (Type) Country Time Period	Type of CS (n)	Timing for Primary Endpoint	aMCS Type (n) Survival (%)	Bleeding (%)/ Severe Bleeding (%) Definition of Bleeding	Transfusion (%)/Number	Limb Ischemia (%)/Severe Limb Ischemia (%) Definition	Other Vascular Complications (%) Definition*
Danial, 2018 Monocentric (retrospective) France 2015-2017	All (814/157 AMICS)	30 d	Percutaneous ECMO (329) 63.8	NR/9.4 Bleeding needing surgical revision after decannulation	NR/NR	8.6/3.0 Requiring surgical revision and/or fasciotomy	14.7 Cannulation adverse events (vessel perforation, arterial dissection, site bleeding) and decanulation complications (surgical revision at time of decanulation for persistent bleeding, or surgical revision after decannulation for false aneurysm, arteriovenous fistula, or septic artery rupture)
			Surgical ECMO (485) 56.3	NR/1.5 Bleeding needing surgical revision after decannulation		12.4/4.5 Requiring surgical revision and/or fasciotomy	3.4 Definition as above
Haurand, 2021 Monocentric (retrospective) Germany 2015-2018	All (106/AMICS 70)	30 d	Impella CP 49.1	16.1/4.8 GUSTO	NR/NR	NR/NR NR	4.8 Peripheral ischemic vascular complications requiring surgical or interventional therapy
Saiydoun, 2022 Monocentric (prospective) France 2018-2021	All (120/51 AMICS)	30 d	Percutaneous ECMO (61) 26.0 Surgical ECMO (59) 17.0	NR/7.0 VARC-2 NR/31.0 VARC-2	NR/NR	NR/3 Requiring percutaneous or surgical intervention or change of cannulation site	NR NR
Choi, 2023 Multicentric (prospective + retrospective) Korea NA	AMICS (315)	30 d	ECMO + culprit-only PCI (175) 48.6 ECMO + MV PCI (140) 60.7	NR/NR NR	NR/NR	NR/NR NR	NR NR
Ikeda, 2023 Multicentric (prospective) Japan 2017-2020	AMICS (593)	30 d	Impella (300) 80.9	NR/5.8 Event in which surgical intervention or transfusion was required	NR/NR	4.1/NR NR	NR NR
			ECMELLA (293) 45.7	NR/9.3 As above		4.7/NR NR	
Drhuva, 2020 Multicentric (retrospective) US 2015-2017	AMICS (28,304)	In-hospital	Medical (16,227) 73.5	NR/11.0 ChestPain-MI registry: decline in hemoglobin level of at least 3 g/ dL; transfusion of whole blood or pRBCs; procedural intervention/surgery at bleeding site to treat bleeding; or documented or suspected retroperitoneal bleed, gastrointestinal bleed, genitourinary bleed, or a bleed in a location not specified elsewhere	NR/NR	NR/NR NR	NR NR
			IABP (8,471) 65.9 Impella (1,768) 55.0 Other pMCS (1,838) NR	NR/16.0 As above NR/31.3 As above NR/NR NR			
	AMICS (3,077)	In-hospital			NR/NR		

(continued on next page)

Table 3 (continued)

Author, Date Site (Type) Country Time Period	Type of CS (n)	Timing for Primary Endpoint	aMCS Type (n) Survival (%)	Bleeding (%)/ Severe Bleeding (%) Definition of Bleeding	Transfusion (%)/Number	Limb Ischemia (%)/Severe Limb Ischemia (%) Definition	Other Vascular Complications (%) Definition*
Miller, 2022 Multicentric (retrospective) US 2015-2020			Impella (817 matched) 59.9	NR/20.3 ICD-9 and ICD-10 = In-patient hospitalization with bleeding or bleeding requiring a transfusion)		NR/NR NR	NR NR
Lemor, 2020 Multicentric (retrospective) US and EU 2015-2017	AMICS (6,290)	In-hospital	IABP (817 matched) 71.7 5.6 Included acute limb ischemia, atheroembolism, vessel dissection, and injury to blood vessels	NR/15.5 As above Impella (5,730/matched 450) 26.7	NR/NR NR	22.2/NR	NR/NR NR
ECMO (560/matched 450) 43.3	31.1/NR	14.4 As above					
Syntila, 2021 Monocentric (retrospective) Germany 2015-2020	AMICS + CA (159)	In-hospital	Impella (105/matched 40) 45.0 ECMO (59/matched 40) 32.5	NR/10.0 Access site bleeding requiring transfusion (≥ 3 pRBCs) NR/32.5 As above	NR/NR	NR/2.5 Requiring intervention NR/20 Requiring intervention	NR NR 1 NR
Lewin, 2023 Monocentric (retrospective) Germany 2017-2021	All (203/75 AMICS)	30 d (severe AEs reported under support)	Impella 5.0 (90) and Impella 5.5 (113) 57.0	NR/22.0 Needing surgical revision	NR/NR	1/NR NR	NR
Almarzoq, 2023 Multicentric (retrospective) US 2015-2019	AMICS (23,478)	30 d death or readmission	Impella (4,063) 46.2 IABP/no MCS (6,964/ 12,451) 61.1	NR/NR NR NR/NR NR	NR/NR NR/NR	NR/NR NR NR/NR NR	NR NR NR NR

Abbreviations: AE, adverse event; aMCS, acute mechanical circulatory support; AMI, acute myocardial infarction; AMICS, acute myocardial infarction cardiogenic shock; BARC, Bleeding Academic Research Consortium; COVID, coronavirus disease; CS, cardiogenic shock; ECMO, extracorporeal membrane oxygenation; ELSO, Extracorporeal Life Support Organization; EU, European Union; GI, gastrointestinal; fFP, frozen fresh plasma; GUSTO, Global Use of Streptokinase and t-PA for Occluded Coronary Arteries; IABP, intra-aortic balloon pump; ICD, implantable cardioverter defibrillator; MCS, mechanical circulatory support; MI, myocardial infarction; MV, multivessel; NR, not reported; PCI, percutaneous coronary intervention; pMCS, percutaneous MCS; pRBCs, packed red blood cells; rRBCs, residual red blood cells; STS ACS, STS Adult Cardiac Surgery Database; TIMI, thrombolysis in myocardial infarction; VA, venoarterial; VV, venovenous; VARC, Valve Academic Research Consortium.

Table 4
Adverse Events in Observational Cardiogenic Shock Studies

Author, Date Site (Type) Country Time Period	SMC (%) Definition	SI (%) Definition	TRB (%), Definition HEM (%), Definition	STK (%) Definition	CPC ≥3	Rec MI (%)/ Device Failure (%) Definition	ICU LOS (d) In-hospital LOS (d)
Danial, 2023 Monocentric (retrospective) France 2015-2018	NR	12	NR NR	11.0 NR	NR	NR/NR	NR NR
Mansour, 2022 Multicentric (retrospective) France 2020-2022	NR	NR	36.4 VV-ECMO, 34.6 ECMO, thrombotic events included ischemic stroke, deep vein thrombosis, pulmonary embolism (or pulmonary thrombosis), acute mesenteric ischemia, acute limb ischemia, macroscopic thrombus of circuit and/ or membrane without needing to change circuit or oxygenator, oxygenator failure requiring change due to clot formation, acute circuit thrombosis requiring change	0.9 VV-ECMO, 9.6 ECMO NR	NR	NR/35.6 VV-ECMO, 21.2 ECMO Circuit clots, circuit change and/or oxygenator failure	28
Schrage, 2023 Multicentric (retrospective) EU and US NR	NR	NR	NR NR 33, LDH ≥ 1,000 U/L and haptoglobin ≤0.3 g/L in 2 consecutive samples within 24 h	11.6 Ischemic stroke, intracerebral bleeding, or hemorrhagic stroke on CT	NR	NR NR	NR NR
Lee, 2021 Multicentric (retrospective) Korea 2014-2018	NR	NR	NRNR	3.4 NR	4.4	NR/ NR	12.7
Acharia, 2020 Multicentric (retrospective) ELSO registry 2014-2018	NR	NR	Circuit clot 8.1%, circuit component clot requiring change of circuit/NR	7.0 Central nervous system infarction and hemorrhage confirmed by CT or MRI	NR	NR/0.7 Change indicated due to equipment failure	NR NR
Schrage, 2023 Multicentric (retrospective) EU and US 2016-2021	NR	NR	NR 15.1, LDH ≥ 1,000 U/L and haptoglobin ≤0.3 g/L in 2 consecutive samples within 24 h NR 1.1, As above	8.6 Intracerebral bleeding/ hemorrhagic or ischemic stroke on CT 4.5 As above	NR	NR/ NR	NR NR
Varshney, 2023 Multicentric (retrospective) EU 2014-2019	NR	NR	NR/ NR	4.0 NR 4.0 NR 6.0 NR 8.0 NR	NR	NR/ NR	NR NR

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Table 4 (continued)

Author, Date Site (Type) Country Time Period	SMC (%) Definition	SI (%) Definition	TRB (%), Definition HEM (%), Definition	STK (%) Definition	CPC ≥ 3	Rec MI (%)/ Device Failure (%) Definition	ICU LOS (d) In-hospital LOS (d)
Danial, 2018 Monocentric (retrospective) France 2015-2017	2.6	16.5 Cannulation site infections were clinically suspected in presence of local (inflammation, cellulitis, purulent drainage) and general signs (fever, hyperleukocytosis) of sepsis; diagnosis of cannulation site infection required positive culture of subcutaneous needle aspirate during VA-ECMO support or within 7 d after VA-ECMO removal	NR NR	NR NR	NR	NR/ NR	NR NR
Haurand, 2021 Monocentric (retrospective) Germany 2015-2018	NR	27.8 As above NR NR	NR NR	NR NR	NR	NR/ NR	8.5
Saiydoun, 2022 Monocentric (prospective) France 2018-2021	NR	2 Groin infections were clinically suspected in presence of local (inflammation, purulent drainage) and general signs of sepsis with positive bacteriological sample	NR NR	NR NR	NR	NR/ NR	10
Choi, 2023 Multicentric 49.3	(prospective + retrospective) Korea NA NR NR	As above NR	NR	NR NR	NR NR	62.9	NR/ NR
Ikeda, 2023 Multicentric (prospective) Japan 2017-2020	NR	NR	1, Based on physicians' reports or specific institutional safety protocols 13, Vent in which patients had increased plasma-free hemoglobin, defined as ≥ 40 mg/dL, within 48 h after Impella support or a clinically significant increase in lactate dehydrogenase, an increase in indirect bilirubin, and a decrease in hemoglobin level within 48 h after Impella support 0.3, As above 8.7, As above	1.0 NR 2.0 NR	NR	NR/ 1.2 Pump stop	NR NR

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Table 4 (continued)

Author, Date Site (Type) Country Time Period	SMC (%) Definition	SI (%) Definition	TRB (%), Definition HEM (%), Definition	STK (%) Definition	CPC ≥ 3	Rec MI (%)/ Device Failure (%) Definition	ICU LOS (d) In-hospital LOS (d)
Drhuva, 2020 Multicentric (retrospective) US 2015-2017	NR	NR	NR NR	NR NR	NR	NR/ NR	NR NR
Miller, 2022 Multicentric (retrospective) US 2015-2020	NR	NR	NR NR	NR NR	NR	NR/ NR	10.3 11.8 9.7 11.7
Lemor, 2020 Multicentric (retrospective) US and EU 2015-2017	NR	NR	NR 1.1, NR NR	1.1 Ischemic stroke 5.6	NR	NR/ NR	NR 7.0 NR
Syntila, 2021 Monocentric (retrospective) Germany 2015-2020	NR	NR	1.1, NR NR NR	Ischemic stroke 0.0 NR	22.2	0/ NR	11.0 NR NR
Lewin, 2023 Monocentric (retrospective) Germany 2017-2021	2.5 Brachial plexus injury defined as nerve damage resulting in residual ipsilateral neurologic dysfunction of upper extremity including neuralgia, paresthesia, or motor deficits	10.9 Prosthetic graft infection definition based on clinical, laboratory, microbiological, and radiological findings	3.4, NR 7.4, NR	NR NR NR	NR	NR/4.9 Damage to purge line, infusion filter, or a sudden decrease in pump flow without signs of thrombosis, requiring immediate explantation $\pm$ replacement according to patient project and myocardial recovery	NR NR
Almarzooq, 2023 Multicentric (retrospective) US 2015-2019	NR NR	NR NR	NR/ NR NR/ NR	NR NR NR NR	NR NR	NR/ NR NR/ NR	NR NR

NOTE. Refer to Table 3 for type of CS, timing for primary endpoint and pMCS type, and number and percent survival for each study.

None of the studies reported events in systemic embolism, structural myocardial injury, stent thrombosis, or urgent revascularization.

Abbreviations: AMICS, acute myocardial infarction cardiogenic shock; BARC, Bleeding Academic Research Consortium; COVID, coronavirus disease; CPC, cerebral performance category; CS, cardiogenic shock; CT, computed tomography; DF, device failure; ECMO, extracorporeal membrane oxygenation; GI, gastrointestinal; HEM, hemolysis; IABP, intra-aortic balloon pump; ICU, intensive care unit; LDH, lactose dehydrogenase; LOS, length of stay; MCS, mechanical circulatory support; MI, myocardial infarction; MRI, magnetic resonance imaging; MV, multivessel; NR, not reported; PCI, percutaneous coronary intervention; pMCS, percutaneous MCS; pRBCs, packed red blood cells; sAE, severe adverse event; SI, site infection; SMC, sensory-motor complications; STK, Stroke; TRB, thrombosis; VA, venoarterial; VARC, Valve Academic Research Consortium; VV, venovenous.

thrombosis or severe vasoconstriction, and the potentially severe management decisions triggered by hemolysis in clinical practice, such as device explantation and/or severe acute kidney injury requiring renal replacement therapy and/or acute thrombosis or severe vasoconstriction, hemolysis remains severely underreported (5 of 16 observational studies and 1 of 7 RCTs), indicating a potential underestimation of its incidence and misconception of its importance. Hemolysis may be related to thrombosis and interfere with the interpretation of some coagulation assays.²³

AE: Thrombosis

Thrombotic complications²⁴ (arterial and venous) have been detailed and reported less in available studies; no RCT and only two observational studies report them as specific secondary endpoints based on investigators' adjudication, signaling another potential underestimation.²⁵

AE: Device Failure

Device failure was reported in only 2 of 4 RCTs and in 3 of 16 observational studies, with very heterogeneous definitions including different complications (clots, decannulation, pump stop for unknown reasons, most frequently requiring device extraction and/or exchange).

AE: Other

Additional AEs are occasionally identified and reported. Few observational studies report recurrent acute myocardial infarction (AMI), urgent revascularization, or stent thrombosis in the AMI with CS (AMICS) population, whereas this is systematically recorded in RCTs on AMICS.^{26,27}

Incidence of AEs in CS Literature

A systematic literature review identified 7 RCTs published on CS since 2012 (Table 2) and 16 large observational studies from 2012 to 2023 comprising 58,950 CS patients, of which 40,346 were admitted with AMICS (Tables 3 and 4). In this group of studies, those after 2015 included >100 patients. Due to the substantial heterogeneity in AE collection and reporting in these studies, gleaning and evaluating the available evidence is challenging and even potentially hazardous.

For example, reported rates of severe bleeding showed high variance and extremely high heterogeneity in the definitions used, with incidences ranging from 3.3% to 36.0% in RCTs and 1.5% to 36.8% in observational studies. Transfusion rates, although rarely reported, range from 22.0% to 31.1% in observational studies and from 30.0% to 46.0% in RCTs. Rates of stroke and severe limb ischemia also varied widely, ranging from 1.0% to 11.6% and 0.9 to 21.0%, respectively. Hemolysis and thrombosis rates in tMCS patients were also disparate across studies at 1.1% to 33.0% and 0.3% to 36.6%, respectively. Device failure rates were difficult to interpret due to the absence of clear definitions and ranged from 0.7% to 21.2%. Access site infection was reported to be between 2.0% and

27.8%. Data from the ELSO Registry suggest a reduction in thrombotic and bleeding HRAEs from 2010 to 2017, but the rate of HRAEs remains significant.²⁸

SHEAR tMCS Stages

Given the variability in definitions and reporting of AEs outlined previously, the proposed SHEAR classification system aims to respond to the urgent need for a clear and standardized description of thrombotic and bleeding phenotypes and events with the specific clinical characteristics of critically ill patients on tMCS. An appropriate stratification can be used to tailor the patient's treatment and standardize the reported outcome in this clinically heterogeneous population.

HRAEs include both hemorrhagic and thrombotic phenomena and are based on common pathophysiological mechanisms as well as on the underlying disease itself. All tMCS patients experience some degree of coagulation activation through blood contact with the tMCS surface or the device's components, which induces thrombogenicity.²⁹ Mechanical and physical forces (eg, heat generation, stagnation of blood, shear forces) also play a role. The hemostatic derangements in critically ill tMCS patients include consumption of platelets and coagulation factors, fragmentation of large molecular weight von Willebrand factor multimers, platelet dysfunction, and reduction of antithrombin levels³⁰ leading to an increased risk for HRAEs. The final clinical picture (ie, the prevalence of thrombotic or bleeding events) is therefore not fully predictable at baseline and may evolve in a highly variable way in each patient.³¹

In this classification, HRAEs are defined as any event that is a direct consequence of the activation and/or derangement of the coagulation process or induced by contact of blood with foreign surfaces. Consideration of biochemical characteristics and clinical manifestations in a pathophysiological relationship to tMCS are also considered. Five stages (labeled A-E) of increasing severity of HRAEs are described, adopting the model of the SCAI CS classification; the different stages rely on three factors: (1) events associated with the device, (2) biochemical laboratory markers, and (3) clinical events.

Events specifically associated with the device are detected by observing the gradual or sudden change in technical parameters or the deterioration or loss of performance of the device. In patients treated with venoarterial extracorporeal membrane oxygenation, for example, an increase in the transmembrane pressure of the oxygenator is an indicator of thrombotic phenomena inside the oxygenator, leading to an increased resistance in blood flow and decreased pressure. On the other hand, in patients on Impella, an increase in the purge fluid pressure indicates an obstacle in the purge system or in the motor itself, but not typically a thromboembolic constellation. The more advanced stages of this classification indicate device-related thrombotic phenomena can cause tMCS malfunction or severe hemolysis, requiring an elective or urgent replacement.

Biochemical characteristics are the markers of hemostatic derangement, which can vary from subtle signs to the overt manifestation of disseminated intravascular coagulation.

increased oxygenator resistance, leading to an increase in the transmembrane pressure. In Impella patients, the increase in the purge fluid pressure can be secondary to microthrombosis at the pump outlet. Furthermore, progressively increased parameters of hemolysis may be observed.

Bleeding. Stage bB patients present a specific laboratory profile, characterized by a decrease of $\geq 50\%$ in the baseline platelet count, accompanied by a decrease in hemoglobin and hematocrit levels while on tMCS. These findings may lead to a transfusion need of ≤ 2 pRBC units/48 h without obvious sources of bleeding.

Stages C, D, and E are characterized by *overt hemocompatibility-related events* with increasing impact on the patient's outcomes and device performance.

Stage C: Clinical HRAEs

Stage C includes uncomplicated HRAEs not leading to a loss of organ function or requiring any intervention other than medical therapy.

Thrombosis. Stage tC HRAEs include uncomplicated DVT, peripheral embolism not requiring intervention and not leading to organ dysfunction, transient ischemic attack, or minor stroke (not treated with systemic thrombolysis or thrombectomy and without permanent neurological consequences), or the presence of an intracardiac thrombus.

Bleeding. Stage bC encompasses access site or surgical site bleeds and gastrointestinal and/or lung bleeds that can be treated conservatively and do not require ≥ 4 pRBC units/48 h.

Stage D: Severe HRAEs

Stage D includes patients experiencing clinical events complicated by partial organ loss of function or requiring intervention other than medical therapy.

Thrombosis. Stage tD thrombotic clinical events include recurrent embolism, arterial embolism resulting in organ dysfunction (eg, acute renal injury), DVT complicated by pulmonary embolism, peripheral thrombosis, or embolism requiring intervention (eg, limb ischemia). Patients with tMCS thrombosis requiring elective device exchange are also included in stage D.

Bleeding. A stage bD event is any bleeding (except for intracranial bleeding) requiring intervention (eg, access site revision or peripheral embolization) or transfusion of ≥ 4 pRBC units/48 h. Patients presenting with severe MCS-related coagulopathy requiring transfusion other than pRBC (platelets and plasma transfusions) and/or cessation of antithrombotic therapy (anticoagulation or antiplatelet therapy) are also included in stage bD.

Stage E: life-threatening HRAEs

Stage E includes life-threatening events with severe or complete organ dysfunction.

Thrombosis. Patients considered tE experience severe thrombotic cardiac-related events like acute MCS failure, acute stent thrombosis and heart valve thrombosis, or extracardiac life-threatening events like bowel ischemia and major stroke.

Bleeding. Similarly, patients are classified as bE when experiencing any life-threatening bleeding leading to acute hemorrhagic shock or any potentially fatal intracranial hemorrhage.

Limitations

This work has some limitations. First, the definition does not consider transfusion protocols (neither thresholds nor strategy), which are currently under-recognized in all existing definition systems aside from SHEAR. Further studies will be required to elucidate their impact on SHEAR staging. On the other hand, introducing a transfusion threshold might further complicate the score, which makes it less valuable for quick bedside assessment. Second, this is a posttreatment evaluation as it ultimately sums up all events during tMCS support. It is not a therapeutic tool but can be leveraged for post-hoc analysis to continuously improve patient care and the design of future clinical trials. Lastly, the assumption that complications are directly related to tMCS is inappropriate, as critically ill patients with CS present multiple reasons for HRAEs beyond tMCS.

Applying this score requires unique attention to a thorough adjudication process to ensure that outcomes are consistently captured and reported. Finally, the score leaves some gray areas and key questions:

- Are four stages enough to reflect the complexity of these patients? As an example, a patient receiving 4 units of pRBCs over 1 week would score the same as a patient who suffers an intracranial hemorrhage. This will need to be further explored soon based on real-life experience.
- Multiple confounders are possible, especially in patients after cardiac surgery (postcardiotomy). For example, hemodilution with low antithrombin levels may contribute to the findings.
- Not all risk factors are equally weighted. For example, patients on aspirin have a much lower risk for bleeding compared with those on triple antithrombotic therapy and/or with low platelet counts.
- The anticoagulant agent used, and the role of heparin-induced thrombocytopenia, are not considered.
- A patient outcome-centered model would allow increased learning and a better-weighted scoring of each category and/or tier.
- The underlying disease might result in severe coagulation disorders (such as disseminated intravascular coagulation), which are not accounted for.

Clinical Vignette

A patient with a history of dilated cardiomyopathy and atrial fibrillation who had a recent percutaneous coronary

intervention with a direct oral anticoagulant and P2Y12-inhibitor as a combination therapy presents with heart failure CS (bA). An axillary Impella 5.5 is inserted and, on day 2, there is an increase in purge fluid pressure and some signs of nonclinical hemolysis (tB, 1 point). On day 3, the patient suffers surgical site bleeding requiring the transfusion of 2 pRBCs (bC, 2 points). On day 5, he develops an AMI due to stent thrombosis (tE, 4 points). The SHEAR score is tB7 (t5-b2).

Conclusions

After a systematic review of existing knowledge and critical and varied discussions, a group of European experts in various areas of tMCS proposes a new hierarchically graded consensus classification for HRAEs for tMCS patients. The resulting aims of this consensus are: (1) to assess the aggregate net burden of HRAEs within and between different tMCSs, (2) to provide an easy bedside tool for further studies, and (3) to increase the awareness of the clinical community of the crucial role of these complications in patient outcomes and the overall health-care burden. The group acknowledges that a prospective validation of this score is needed, and strongly urge researchers to begin reporting AEs according to this SHEAR definition to achieve consistent validation of investigations in device-stratified outcomes and the field of CS.

Declaration of competing interest

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